# First- and Second-Order Optical Coherence as Probes of Spacetime Foam

A. G. Kim*
*Physics Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA*

P. E. Nugent
*Applied Mathematics and Computational Research Division,*
*Lawrence Berkeley National Laboratory Berkeley, California 94720, USA*

L. Wang
*Department of Physics and Astronomy 4242 TAMU College Station, Texas 77843-4242, USA*
(Dated: May 4, 2026)

Quantum gravity is expected to render spacetime stochastic at the Planck scale. We show that the resulting random arrival-time jitter affects the field-correlation and intensity-correlation functions of light from distant sources differently, breaking the Gaussian-statistics identity (the Siegert relation) that links them for all chaotic-thermal light. For spatially extended sources, foam-induced jitter between rays from different parts of the source produces an additional suppression of the zero-lag photon-bunching (Hanbury Brown–Twiss) peak that depends on the source angular size and probes the transverse coherence length of the foam. Both signatures are achromatic and follow a distance scaling distinct from plasma scintillation or atmospheric turbulence, making them complementary to the existing constraints from imaging and from energy-dependent time-of-flight measurements. We develop the framework in the narrow-band, weak-jitter, point-source limit and its extension to spatially resolved sources, recover the published Strehl-ratio and fringe-visibility tests as limiting cases, and quantify the predicted observables for the Ng–Perlman family of foam models. For the holographic exponent ($\alpha = 2/3$), the framework sets upper limits on the line-of-sight foam path-length variance and on the transverse coherence length that approach but do not clearly exceed the systematic floors anticipated for next-generation optical intensity interferometers.

## I. INTRODUCTION

A long-standing expectation from the union of quantum mechanics and general relativity is that the smooth pseudo-Riemannian manifold of classical spacetime breaks down at the Planck length $\ell_{\rm P} = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m. In Wheeler's original "spacetime foam" picture [1], quantum fluctuations of the metric on this scale render the geometry stochastic; similar conclusions are reached in the Euclidean path-integral treatment [2] and in modern holographic and entropic-gravity arguments from very different starting points. While a complete theory of quantum gravity remains elusive, the question of whether residual imprints of Planck-scale geometric fluctuations can accumulate over cosmological baselines and become observable is one of the few empirical handles on physics in the Planck regime, and it has motivated a substantial body of phenomenological work over the past two decades [3, 4].

A widely studied parametrization, due to Ng and collaborators [4–6], assigns the cumulative line-of-sight path-length variance acquired by light traversing a distance $D$ the power-law form

$$\sigma_\ell^2(D) \;=\; A_\alpha\, D^{2(1-\alpha)}\, \ell_{\rm P}^{2\alpha}, \tag{1}$$

where $A_\alpha$ is a dimensionless coefficient of order unity and the exponent $\alpha$ encodes the underlying foam dynamics. Three values are conventionally distinguished. The "conservative" case $\alpha = 1$ follows from Wheeler's original dimensional argument and gives $\sigma_\ell \sim \ell_{\rm P}$ regardless of $D$ – observationally inconsequential. The "holographic" case $\alpha = 2/3$ is motivated by the requirement that the information content of a region of spacetime saturate the Bekenstein–Hawking bound and scale with the area of its boundary in Planck units [5]; it predicts $\sigma_\ell \sim (D\ell_{\rm P}^2)^{1/3}$, of order $10^{-15}$ m over 1 Gpc. The "random-walk" case $\alpha = 1/2$ treats successive Planck cells as independent path-length kicks and gives $\sigma_\ell \sim (D\ell_{\rm P})^{1/2}$, roughly $10^{-5}$ m over the same distance – large enough to randomize the optical phase of light from sources at $z \gtrsim 1$. The three cases bracket the range investigated most thoroughly; for a recent review situating them within the broader quantum-gravity phenomenology program, see Carlip [3].

The foam line of inquiry sits within a broader program of quantum-gravity phenomenology that has matured substantially in the multi-messenger era [7, 8]. Other channels covered in those reviews include searches for Lorentz-invariance violation in high-energy astrophysical transients, vacuum birefringence over cosmological baselines, and

* agkim@lbl.gov

modifications of gravitational-wave propagation; complementary laboratory programs target Planck-scale metric noise via tabletop interferometers [9], and a parallel theoretical line asks whether quantum-gravitational degrees of freedom decohere otherwise classical photon states during propagation [10]. The present work belongs to the propagation-based, coherence-channel branch of this landscape: we ask what imprint the Ng–Perlman foam ansatz leaves on the first- and second-order temporal correlations of light from cosmologically distant sources.

Two complementary observational channels have been used to test these models. The first exploits the fact that random Planck-scale path-length deviations imprint a stochastic phase $\sigma_\phi = 2\pi\sigma_\ell/\lambda$ across the aperture of any telescope, reducing the Strehl ratio by $S \approx \exp(-\sigma_\phi^2)$ and degrading the diffraction-limited point-spread function. Detections of crisp, near-diffraction-limited images of distant active galactic nuclei with the Hubble Space Telescope and ground-based instruments [11–14], together with stellar interferometry, exclude $\alpha \lesssim 2/3$ at optical wavelengths for $D \sim 1$–10 Gpc, leaving the holographic and conservative cases as the only quantitatively viable scalings. These exclusions assume transverse foam correlations are negligible on aperture scales; models with long-range transverse correlations can evade blurring constraints while retaining large $\sigma_\ell(D)$ [15]. The second channel uses energy-dependent time-of-flight measurements of high-energy astrophysical transients: gamma-ray bursts observed by Fermi [16] and TeV blazar flares observed by H.E.S.S. [17] constrain the linear and quadratic dispersion that some quantum-gravity models predict, with LHAASO observations of GRB 221009A [18] providing the current leading limits. Together the two channels have winnowed the parameter space but not eliminated it, and several proposed scalings – including the holographic $\alpha = 2/3$ – remain unconstrained at any wavelength accessible to current instrumentation. A recent analysis of GRB 221009A claims consistency with holographic foam blurring [19], though without formal statistical significance.

Both existing channels share a structural limitation that motivates the present work. The imaging tests are essentially binary: a wavefront either remains coherent enough to image or it does not, so that once a positive detection is registered the constraint saturates and no further information is recovered about the magnitude of the residual phase variance. The time-of-flight tests, conversely, rely on a chromatic dispersion signature that is precisely what is *not* expected from a geometric-optics path-length fluctuation, for which the timing offset $\delta t = \Delta\ell/c$ is frequency-independent. The Fermi, H.E.S.S., and LHAASO analyses specifically seek a Lorentz-invariance-violating (LIV) dispersion of the form $\delta t \propto E^n/E_{\rm QG}^n$: a deterministic, energy-ordered delay. A stochastic, energy-independent geometric jitter produces no such signal, so the absence of a detected LIV dispersion in GRB or blazar data leaves the Ng–Perlman foam entirely unconstrained by those measurements. A foam model whose accumulated phase variance lies below the imaging threshold but whose timing jitter is achromatic is therefore invisible to both channels. Whether observables exist that bridge this gap – sensitive to a phase variance too small to erase fringes outright, but achromatic and thus distinguishable from plasma scintillation or instrumental dispersion – is a natural question, and it is the question we address here.

We develop such a framework based on the first-order (field–field) and second-order (intensity–intensity) coherence functions of light from astrophysical sources, denoted $g^{(1)}(\tau)$ and $g^{(2)}(\tau)$ in standard quantum-optics notation. The first-order coherence is equivalently the fringe visibility measured in a stellar interferometer at delay $\tau$; the second-order coherence is the intensity correlation measured in a Hanbury Brown–Twiss experiment [20] (intensity correlations between two photon-counting detectors, originally developed for stellar angular-diameter measurements). For all stationary chaotic-thermal sources the two are linked by the Siegert relation [21], the Gaussian-statistics identity $g^{(2)}(\tau) - 1 = |g^{(1)}(\tau)|^2$. The framework recovers the published imaging and fringe-visibility tests as limiting cases of a single first-order observable, and we identify two additional foam signatures inaccessible to those tests: one accessible to single-aperture temporal coherence measurements on point-like sources, and one to intensity interferometry on spatially resolved sources such as that recently demonstrated by VERITAS [22, 23]. Their derivation, observability, and use as discriminants against conventional propagation effects are the subject of this paper.

The remainder of the paper is organized as follows. Section II develops the model: the spatiotemporal covariance of the foam-induced path-length fluctuation field, the differential-jitter distribution at a single ray, and its generalization to ray pairs. Section III treats the first-order coherence and its connection to the published Strehl-ratio and fringe-visibility constraints. Section IV treats the second-order coherence, including its conditional and unconditional forms in the narrow-band limit. Section V examines the Siegert relation under the foam-modified field statistics. Section VI extends the analysis to spatially resolved sources and discusses the observability of the resulting signatures, the distance scaling, and the achromaticity as discriminants against conventional propagation phase screens. Section VII concludes.

## II. MODEL

### A. Arrival-Time Jitter as a Time Reparameterization

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\rm obs}(t) = E_{\rm src}(t - \delta t(t))\,, \tag{2}$$

where $E_{\rm src}(t)$ is a stationary zero-mean complex Gaussian process (a single chaotic/thermal mode) and $\delta t(t)$ is the foam-induced arrival-time jitter, a stationary stochastic process statistically independent of $E_{\rm src}$. The jitter arises from cumulative path-length fluctuations along the line of sight of propagation distance $D$:

$$\delta t(t) = \frac{1}{c}\,\Delta\ell_{\rm tot}(t). \tag{3}$$

In the geometric-optics limit the path-length shift $\Delta\ell_{\rm tot}$ is a purely geometric quantity and hence frequency-independent (achromatic), in contrast to plasma dispersion which scales as $\lambda^2$. This achromaticity is a key discriminant between foam and conventional propagation effects.

### B. Spatial and Temporal Fluctuations of Spacetime Foam

Following Ng and collaborators [4–6] (see also Ref. [3] for a review of foam models and their observational constraints), the single-ray path-length variance obeys the power-law scaling

$$\sigma_\ell^2(D) = A_\alpha\, D^{2(1-\alpha)}\,\ell_{\rm P}^{2\alpha}, \tag{4}$$

where $\alpha$ parametrizes the foam model ($\alpha = 1$: conservative; $\alpha = 2/3$: holographic; $\alpha = 1/2$: random walk), and $A_\alpha$ is a dimensionless coefficient of order unity.

Because foam is a property of spacetime rather than of any preferred direction, the same dimensional logic that governs longitudinal path-length fluctuations applies equally to transverse baselines. For two rays separated by a transverse distance $b$, the Ng–Perlman scaling gives a characteristic transverse coherence length

$$r_\perp \sim A_\alpha^{1/2}\, b^{1-\alpha}\,\ell_{\rm P}^{\alpha}, \tag{5}$$

by dimensional analogy with Eq. (4). This analogy motivates the *existence* of a transverse scale but does not derive its value: in the longitudinal case $D$ counts the number of cells traversed along the ray, while in the transverse case $b$ is a separation between rays that share cells along the line of sight  the number of shared cells depends on both $b$ and $D$ in a way that is geometrically distinct from the single-ray accumulation. Accordingly, $r_\perp$ is treated as an independent free parameter in Section VI. We denote the normalized transverse coherence function by $\Phi(b/r_\perp)$, satisfying $\Phi(0) = 1$ and $\Phi(b/r_\perp) \to 0$ for $b \gg r_\perp$, and treat it as a free function with $r_\perp$ constrained empirically by the angular-size dependence of $\delta g^{(2)}$ (Section VI); its model-dependence is discussed below.

Turning to the temporal dimension, quantum foam models posit that each Planck-scale cell of spacetime along the line of sight fluctuates in time. Two photons traversing the same cell at coordinate times separated by $\tau$ experience path-length fluctuations that are correlated for $\tau \ll \tau_{\rm foam}$ but decorrelated for $\tau \gg \tau_{\rm foam}$, where $\tau_{\rm foam}$ is the characteristic temporal fluctuation timescale at each cell. We introduce $C_\alpha(\tau; D)$ as the normalized *temporal* autocorrelation of the per-cell path-length fluctuation, integrated over the $\sim D/\ell_{\rm P}$ cells along the line of sight (where $\ell_{\rm P} = \sqrt{\hbar G/c^3} \simeq 1.616\times10^{-35}$ m is the Planck length). The timescale $\tau_{\rm foam}$ is an additional free parameter of the foam model, not fixed by the Ng–Perlman variance relation alone. If $\tau_{\rm foam}$ is of order the Planck time, $\ell_{\rm P}/c \approx 5\times10^{-44}$ s, then $C_\alpha(\tau) \approx 0$ for all accessible lags: the foam is fully decorrelated on any observable timescale, the most optimistic scenario for detection. In the opposite *quasi-static* limit $\tau_{\rm foam} \gg D/c$ (foam coherent across the entire line of sight), $C_\alpha(\tau) \approx 1$ for all $\tau$ of interest and every temporal observable vanishes.

All numerical estimates in Sections III–VI are evaluated in the dynamical limit $C_\alpha \to 0$, which maximizes the predicted signal at any given lag; the computed signals are therefore conditional upper bounds, not predictions of a fully specified foam model. The parameters $A_\alpha$, $C_\alpha(\tau; D)$, $\tau_{\rm foam}$, $r_\perp$, and $\Phi_{\rm eff}$ are convenient parameterizations of distinct physical effects — single-ray variance, temporal foam dynamics, transverse coherence, and source angular structure — and are not claimed to be simultaneously determined by any single observation.

The Ng–Perlman variance scaling Eq. (4) does not pin the functional forms of $C_\alpha(\tau; D)$ and $\Phi(b/r_\perp)$, and specific foam models predict different shapes. The pixellon model [15] gives a transverse structure with long-range angular

components that does not factorize as $\Phi(b/r_\perp)$ (Legendre polynomial series in aperture-plane angle, with the long-range character strongly suppressing image blurring relative to the uncorrelated case); holographic interferometric proposals in the Verlinde–Zurek class [9] tie the metric-fluctuation noise spectrum to the geometry of the propagation region through the holographic principle, implying model-specific temporal correlation structure. We keep $C_\alpha$ and $\Phi$ as free functions; quantitative comparison with specific model-predicted kernels would require evaluating the convolution integrals of Sections III–VI with those shapes, which we leave to future work.

The full spatiotemporal covariance between two rays separated by transverse distance $b$ takes the factorized form

$$\big\langle \Delta\ell_{\rm tot}(t;\boldsymbol{\theta})\,\Delta\ell_{\rm tot}(t+\tau;\boldsymbol{\theta}')\big\rangle = \sigma_\ell^2(D)\,C_\alpha(\tau;D)\,\Phi\!\left(\frac{b(\boldsymbol{\theta},\boldsymbol{\theta}')}{r_\perp}\right), \tag{6}$$

where $b(\boldsymbol{\theta},\boldsymbol{\theta}') \simeq D|\boldsymbol{\theta}-\boldsymbol{\theta}'|$. The single-ray temporal covariance appropriate for a point source is the special case $\boldsymbol{\theta}=\boldsymbol{\theta}'$ (so $b=0$):

$$\big\langle \Delta\ell_{\rm tot}(t)\,\Delta\ell_{\rm tot}(t+\tau)\big\rangle = \sigma_\ell^2(D)\,C_\alpha(\tau;D). \tag{7}$$

This product form separating the total variance from the normalized spatiotemporal shape is a structural assumption of the model, not a consequence of Eq. (4); in a general stationary process the variance and the normalized correlation functions can carry independent $D$-dependence. The factorization defines a class of foam models within which the analytic framework is exact, analogous to the use of factorized kernels in turbulence phenomenology. Violations of factorization could in principle be detected as a failure of the predicted joint $D$-scaling between $\mathcal{S}(\tau)$ and $\delta g^{(2)}$; we are not aware of a specific quantum-gravity calculation that either confirms or refutes this ansatz. The two principal observables developed below — the Siegert discrepancy $\mathcal{S}(\tau)$ and the zero-lag bunching suppression $\delta g^{(2)}$ — therefore both descend from the single scaling relation Eq. (4), with Sections III–V probing its longitudinal projection and Section VI its transverse projection.

### C. Differential Jitter and Its Distribution

With the spatiotemporal covariance model in place, we define the key derived quantity that enters the coherence integrals. Define the *cross-ray differential jitter* between a photon arriving along direction $\boldsymbol{\theta}$ at time $t+\tau$ and a photon arriving along $\boldsymbol{\theta}'$ at time $t$:

$$\Delta_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\tau) \equiv \delta t(t+\tau;\boldsymbol{\theta}) - \delta t(t;\boldsymbol{\theta}'). \tag{8}$$

For fixed $D$, stationarity implies the statistics depend only on $\tau$ and the transverse separation $b(\boldsymbol{\theta},\boldsymbol{\theta}') = D|\boldsymbol{\theta}-\boldsymbol{\theta}'|$. Expanding the square and applying Eq. (6):

$$\begin{aligned}\sigma_\Delta^2(\tau;\boldsymbol{\theta},\boldsymbol{\theta}') &= \frac{1}{c^2}\big\langle[\Delta\ell_{\rm tot}(t+\tau;\boldsymbol{\theta}) - \Delta\ell_{\rm tot}(t;\boldsymbol{\theta}')]^2\big\rangle \\ &= \frac{1}{c^2}\left[\sigma_\ell^2 + \sigma_\ell^2 - 2\sigma_\ell^2\,C_\alpha(\tau)\,\Phi\!\left(\frac{b}{r_\perp}\right)\right] \\ &= \frac{2\sigma_\ell^2(D)}{c^2}\left[1 - C_\alpha(\tau;D)\,\Phi\!\left(\frac{b(\boldsymbol{\theta},\boldsymbol{\theta}')}{r_\perp}\right)\right].\end{aligned} \tag{9}$$

For $b \gg r_\perp$, $\Phi \to 0$ and the variance becomes $2\sigma_\ell^2/c^2$ *at all lags, including* $\tau=0$ a new effect absent in the point-source case.

By the central limit theorem, $\Delta_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\tau)$ is a sum of contributions from many foam cells and is therefore Gaussian with zero mean:

$$\Delta_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\tau) \sim \mathcal{N}\big(0,\,\sigma_\Delta^2(\tau;\boldsymbol{\theta},\boldsymbol{\theta}')\big), \tag{10}$$

where $\sigma_\Delta^2$ is taken from Eq. (9).

The Gaussian approximation requires the elementary foam cells to be approximately independent. For the cross-ray case the two rays share foam cells along the line of sight, and independence requires $b \gg r_\perp$. In the intermediate regime $b \sim r_\perp$ (equivalently $r_\perp \sim D\Delta\theta$), the cross-ray Gaussian approximation and the $\Phi_{\rm eff}$ approximation of Section VI D break down together; both introduce errors of the same order and must be assessed jointly.

## III. FIRST-ORDER COHERENCE

In Sections III–V we restrict to a point source ($b = 0$), for which the single-ray differential jitter variance reduces to Eq. (9) with $\Phi(0) = 1$:

$$\sigma_\Delta^2(\tau; D) = \frac{2\sigma_\ell^2(D)}{c^2}\big[1 - C_\alpha(\tau; D)\big]. \tag{11}$$

### A. Jitter Convolution Formula for $g^{(1)}$

The observed first-order coherence is

$$g^{(1)}_{\rm obs}(\tau) = \frac{\langle E^*_{\rm obs}(t)\, E_{\rm obs}(t+\tau)\rangle}{\langle |E_{\rm obs}(t)|^2\rangle}. \tag{12}$$

Substituting Eq. (2) and letting $u = t - \delta t(t)$, $v = t + \tau - \delta t(t+\tau)$, so that $v - u = \tau - \Delta(\tau)$, stationarity of $E_{\rm src}$ gives

$$\langle E^*_{\rm src}(u)\, E_{\rm src}(v)\rangle = \langle |E_{\rm src}|^2\rangle\, g^{(1)}_{\rm src}(\tau - \Delta(\tau)). \tag{13}$$

Since $\delta t$ is independent of $E_{\rm src}$, averaging over the jitter yields

$$g^{(1)}_{\rm obs}(\tau) = \big\langle g^{(1)}_{\rm src}(\tau - \Delta(\tau))\big\rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_\Delta^2(\tau))\, g^{(1)}_{\rm src}(\tau - \Delta)\, d\Delta. \tag{14}$$

The observed first-order coherence is the intrinsic coherence function convolved with the Gaussian differential-jitter kernel.

### B. Closed Forms for Lorentzian and Voigt Sources

As a representative case admitting a closed-form result, we take a Lorentzian line profile, which corresponds to an exponential coherence function via the Wiener–Khinchin theorem:

$$g^{(1)}_{\rm src}(\tau) = e^{-|\tau|/\tau_c}, \tag{15}$$

where $\tau_c$ is the coherence time. Evaluating Eq. (14) by completing the square and splitting on the sign of $\tau - \Delta$ gives, for $\tau > 0$:

$$g^{(1)}_{\rm obs}(\tau) = e^{\sigma_\Delta^2/2\tau_c^2} \cdot \frac{1}{2}\left[e^{-\tau/\tau_c}\,\mathrm{erfc}\!\left(\frac{\sigma_\Delta/\tau_c - \tau/\sigma_\Delta}{\sqrt{2}}\right) + e^{\tau/\tau_c}\,\mathrm{erfc}\!\left(\frac{\sigma_\Delta/\tau_c + \tau/\sigma_\Delta}{\sqrt{2}}\right)\right]. \tag{16}$$

In the weak-jitter limit $\sigma_\Delta \ll \tau_c$, the Gaussian average of $e^{-|\tau-\Delta|/\tau_c}$ over $\Delta \sim \mathcal{N}(0, \sigma_\Delta^2)$ is dominated by $|\Delta| \ll |\tau|$, so one expands around $\Delta = 0$:

$$g^{(1)}_{\rm obs}(\tau) \approx e^{-|\tau|/\tau_c}\left[1 + \frac{\sigma_\Delta^2(\tau)}{2\tau_c^2} + \mathcal{O}(\sigma_\Delta^4)\right]. \tag{17}$$

Here $\sigma_\Delta \equiv \sigma_\Delta(\tau)$ is $\tau$-dependent through $C_\alpha(\tau)$; the expansion is in $\sigma_\Delta(\tau)/\tau_c$ at fixed $\tau$, and its validity requires $\sigma_\Delta(\tau) \ll \tau_c$ at each lag of interest. The correction is positive because $e^{-|\tau|/\tau_c}$ is convex and Jensen's inequality applies. The normalization $g^{(1)}_{\rm obs}(0) = 1$ is preserved exactly.

For a Voigt profile $g^{(1)}_{\rm src}(\tau) = e^{-|\tau|/\tau_L}\, e^{-\tau^2/2\tau_G^2}$ — a Lorentzian of width $\tau_L$ convolved with a Gaussian of width $\tau_G$ modelling Doppler and turbulent broadening — the jitter kernel combines with the source Gaussian into an effective width and mean,

$$\tilde\sigma_1^2 = \frac{\sigma_\Delta^2\,\tau_G^2}{\sigma_\Delta^2 + \tau_G^2}, \qquad \mu_1(\tau) = \frac{\tau_G^2\,\tau}{\sigma_\Delta^2 + \tau_G^2}, \tag{18}$$

giving, for $\tau > 0$:

$$g^{(1)}_{\rm obs}(\tau) = \frac{\tilde\sigma_1}{\sigma_\Delta}\, e^{-\tau^2/2(\sigma_\Delta^2+\tau_G^2)}\, e^{\tilde\sigma_1^2/2\tau_L^2}\, \frac{1}{2}\left[e^{-\mu_1/\tau_L}\,\mathrm{erfc}\!\left(\frac{\tilde\sigma_1/\tau_L - \mu_1/\tilde\sigma_1}{\sqrt{2}}\right) + e^{\mu_1/\tau_L}\,\mathrm{erfc}\!\left(\frac{\tilde\sigma_1/\tau_L + \mu_1/\tilde\sigma_1}{\sqrt{2}}\right)\right]. \tag{19}$$

The normalization $g^{(1)}_{\rm obs}(0) = 1$ is preserved: as $\sigma_\Delta \to 0$, $\tilde{\sigma}_1/\sigma_\Delta \to 1$, the exponential prefactors approach unity, and the erfc combination gives 1. In the limit $\tau_G \to \infty$ (pure Lorentzian), $\tilde{\sigma}_1 \to \sigma_\Delta$ and $\mu_1 \to \tau$, so Eq. (19) reduces to Eq. (16). In the limit $\tau_L \to \infty$ (pure Gaussian), the erfc terms combine to unity and $g^{(1)}_{\rm obs}(\tau) = (\tilde{\sigma}_1/\sigma_\Delta)\, e^{-\tau^2/2(\sigma_\Delta^2+\tau_G^2)}$: the observed coherence is a Gaussian whose width is broadened in quadrature by the jitter.

### C. Narrow-Band Limit

For a quasi-monochromatic source with center frequency $\omega_0$ and angular-frequency bandwidth $\Delta\omega \ll \omega_0$, write $E_{\rm src}(t) = A(t)\, e^{-i\omega_0 t}$ where $A(t)$ varies on the coherence timescale $\tau_c = 1/\Delta\omega = \lambda^2/(2\pi c\,\Delta\lambda)$. When $\sigma_\ell/(c\tau_c) \ll 1$, typical realizations of $\delta t$ leave the envelope unaffected, $A(t-\delta t) \approx A(t)$, and the jitter acts purely as a carrier phase modulation; the observed field takes the form

$$E_{\rm obs}(t) \approx A(t)\, e^{-i\omega_0 t}\, e^{i\omega_0 \delta t(t)}. \tag{20}$$

(Corrections to the envelope approximation are of order $(\sigma_\ell/c\tau_c)^2$; for $\alpha = 1/2$ at $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$ this parameter is $\approx 0.28$, so the exact convolution Eq. (16) should be used instead. For $\alpha = 2/3$ the analogous parameter is $\sim 10^{-11}$ (Table I), so Eq. (22) is exact to any practical precision in the holographic case.) Since $A(t)$ and $\delta t(t)$ are independent, the first-order coherence factorizes. The jitter contribution is the characteristic function of $\Delta(\tau)$ evaluated at $\omega_0$; since $\Delta(\tau)$ is Gaussian (Section II C),

$$\left\langle e^{i\omega_0 \Delta(\tau)} \right\rangle_{\delta t} = e^{-\omega_0^2 \sigma_\Delta^2(\tau)/2}. \tag{21}$$

This holds when $\tau \gg \tau_{\rm foam}$ so that foam-cell contributions are independent; in the dynamical limit $\tau_{\rm foam} \ll \tau_c$ the condition is satisfied at all observationally relevant lags. Substituting $\omega_0 = 2\pi c/\lambda$ and Eq. (11):

$$g^{(1)}_{\rm obs}(\tau) = g^{(1)}_{\rm src}(\tau) \cdot \exp\!\left(-\sigma_\phi^2 \left[1 - C_\alpha(\tau; D)\right]\right), \tag{22}$$

where we define the rms accumulated phase

$$\sigma_\phi \equiv \frac{2\pi\, \sigma_\ell(D)}{\lambda} = \frac{2\pi \sqrt{A_\alpha}\, D^{1-\alpha}\, \ell_{\rm P}^{\alpha}}{\lambda}. \tag{23}$$

We write the foam suppression factor as

$$\mathcal{F}_{\rm foam}(\tau) \equiv e^{-\sigma_\phi^2 [1 - C_\alpha(\tau; D)]}, \tag{24}$$

so $g^{(1)}_{\rm obs}(\tau) = g^{(1)}_{\rm src}(\tau) \cdot \mathcal{F}_{\rm foam}(\tau)$. At $\tau = 0$: $\mathcal{F}_{\rm foam}(0) = 1$. At $\tau \to \infty$: $\mathcal{F}_{\rm foam}(\infty) = e^{-\sigma_\phi^2 [1 - C_\alpha(\infty; D)]}$, which equals $e^{-\sigma_\phi^2}$ in the dynamical limit ($C_\alpha \to 0$) and approaches unity for quasi-static foam ($C_\alpha \approx 1$).

### D. Regime Summary

Eq. (14) is always valid (exact within the model assumptions); Eq. (16) is its closed form for a Lorentzian source and is likewise exact. Eq. (17) (weak-jitter expansion) requires $\sigma_\Delta(\tau)/\tau_c \ll 1$ at each lag of interest. Eq. (22) (narrow-band, carrier-phase result) requires $\sigma_\ell/(c\tau_c) \ll 1$; it captures the dominant physics when $\sigma_\phi \gtrsim 1$ but $\sigma_\phi \ll \lambda/\Delta\lambda$. These conditions are compatible: the narrow-band condition controls whether the amplitude envelope is displaced by the jitter, while the weak-jitter condition controls whether the carrier-phase suppression is small. When neither approximation applies (i.e. $\sigma_\ell/(c\tau_c) \sim 1$), the full convolution Eq. (14) must be evaluated.

### E. Connection to Fringe Visibility and Strehl Ratio

The fringe visibility and Strehl ratio are the observables through which published constraints on spacetime foam have been obtained [4, 11–14]. Both are direct consequences of the first-order coherence $g^{(1)}_{\rm obs}(\tau)$ derived above, and the foam suppression factor $\mathcal{F}_{\rm foam}(\tau)$ of Eq. (24) unifies them in a single framework.

### 1. Fringe Visibility and the Lieu–Hillman Condition

The fringe visibility measured in a stellar interferometer at time delay $\tau$ is $V(\tau) = |g^{(1)}_{\rm obs}(\tau)|$. From Eq. (22):

$$V(\tau) = V_{\rm src}(\tau) \cdot \mathcal{F}_{\rm foam}(\tau), \tag{25}$$

where $V_{\rm src}(\tau) = |g^{(1)}_{\rm src}(\tau)|$ is the intrinsic source visibility. Lieu & Hillman [11] argue that fringes are erased when the accumulated phase randomizes over $[0, 2\pi)$, i.e. $\sigma_\phi \gtrsim 1$ — the regime in which the rms accumulated phase exceeds one radian and randomizes the phase across the aperture, erasing interferometric fringes. In our language this is the limit $\mathcal{F}_{\rm foam} \to 0$: the foam suppression factor collapses and $V \to 0$. The Lieu–Hillman condition $\Delta\phi \geq 2\pi$, where $\Delta\phi \simeq 2\pi\sigma_\ell(D)/\lambda$ [4], is precisely $\sigma_\phi \geq 1$. Our result extends this binary criterion to a continuous suppression curve $V(\tau)$ whose shape encodes the temporal decorrelation $C_\alpha(\tau; D)$.

### 2. Strehl Ratio

The Strehl ratio $S$ the ratio of the observed PSF peak to the diffraction-limited peak is the standard measure of wavefront quality in telescope optics. In the Ng–Perlman picture [4], spacetime foam imprints a spatially varying random phase $\Delta\phi(x, y)$ across the entrance aperture: each point $(x, y)$ on the mirror receives a ray that has accumulated an independent random path-length shift from the Planck-scale cells it traversed, producing a random phase screen with rms value $\sigma_\phi$. When these phase shifts are spatially uncorrelated across the aperture the limit $r_\perp \ll D_{\rm tel}$, where $r_\perp$ is the transverse foam coherence length and $D_{\rm tel}$ is the aperture diameter the Maréchal approximation (the small-phase-variance expansion of the diffraction-limited PSF reduction) gives

$$S = e^{-\sigma_\phi^2}, \tag{26}$$

valid for $\sigma_\phi \lesssim 1$; for $\sigma_\phi \gg 1$ the PSF peak is effectively destroyed.

Within the present temporal framework, the same quantity $\sigma_\phi^2$ appears as the large-lag limit of the foam suppression factor. From Eq. (24), $\mathcal{F}_{\rm foam}(\tau) = e^{-\sigma_\phi^2[1-C_\alpha(\tau;D)]}$, so in the dynamical limit $C_\alpha(\infty; D) \to 0$:

$$S = \mathcal{F}_{\rm foam}(\tau \to \infty) = e^{-\sigma_\phi^2}. \tag{27}$$

This equality is an exact result of the Gaussian characteristic function, independent of the Maréchal approximation, but holds only in the dynamical limit $C_\alpha(\infty; D) \to 0$; for foam that does not fully decorrelate on any accessible timescale the two sides differ. It reflects a correspondence between two descriptions of the same underlying physics: the spatial description sums independent phase contributions across the mirror at a single instant, while the temporal description tracks how the phase along a single ray decorrelates as the lag $\tau$ grows large enough that $C_\alpha(\tau; D) \to 0$ and the two time samples become independent. Both involve the same rms phase variance $\sigma_\phi^2 = (2\pi\sigma_\ell/\lambda)^2$, but the correspondence is asymmetric: the temporal result $\mathcal{F}_{\rm foam}(\tau \to \infty) = e^{-\sigma_\phi^2}$ is exact for all $\sigma_\phi$, while the spatial Marchal formula is an approximation valid only for $\sigma_\phi \lesssim 1$. For $\sigma_\phi \gg 1$ both give $S \approx 0$, so they agree numerically, but only the temporal result is theoretically exact in that regime. For quasi-static foam ($C_\alpha \approx 1$), $\mathcal{F}_{\rm foam}(\infty) \to 1$ consistently with $S \to 1$ in the $r_\perp \gg D_{\rm tel}$ limit discussed below.

It is important to distinguish the role of the transverse coherence function $\Phi(b/r_\perp)$ from the aperture-plane phase noise just described. The quantity $\sigma_\phi^2$ is a single-ray variance: it characterises the accumulated phase fluctuation along one line of sight and does not by itself involve correlations between spatially separated rays. The Marchal formula Eq. (26), however, additionally requires that adjacent aperture elements sample *independent* foam realizations, which is precisely the $r_\perp \ll D_{\rm tel}$ condition already stated. In the opposite limit, $r_\perp \gg D_{\rm tel}$, the entire aperture sits within a single coherence patch: the wavefront has a common phase offset that varies in time but not across the aperture, there is no spatial phase variation, and $S \to 1$ the foam is invisible to PSF-based measurements even though $\sigma_\phi^2$ may be large. The transverse coherence function $\Phi(b/r_\perp)$ governs a physically distinct effect correlations between rays arriving from different sky directions $\boldsymbol{\theta} \neq \boldsymbol{\theta}'$ which manifests in the second-order coherence of extended sources (Section VI) but does not directly enter the single-aperture Strehl ratio.

Both the Strehl and the Lieu–Hillman fringe visibility therefore measure the same underlying quantity $\sigma_\phi^2 = (2\pi\sigma_\ell/\lambda)^2$. The difference is observational channel: Strehl and fringes probe the *spatial* phase variation across the aperture (requiring $r_\perp \ll D_{\rm tel}$), while the temporal $g^{(1)}$ of Eq. (22) probes the *temporal* phase variation along a single ray as a function of lag, with no assumption on $r_\perp$ relative to the aperture.

## IV. SECOND-ORDER COHERENCE

We derive the second-order coherence $g^{(2)}_{\rm obs}(\tau)$ for a point source, continuing the framework of Section III. The result has the same structure as Eq. (14): $g^{(2)}_{\rm obs} - 1$ is a convolution of $g^{(2)}_{\rm src} - 1$ with the identical Gaussian jitter kernel. Because the intensity $|E_{\rm obs}|^2$ is blind to the carrier phase $e^{i\omega_0\delta t}$, foam leaves $g^{(2)}$ nearly unchanged in the narrow-band limit while strongly suppressing $g^{(1)}$ — the asymmetry that produces the Siegert violation developed in Section V.

### A. Jitter Convolution Formula for $g^{(2)}$

Fix a realization of the jitter process $\delta t(\cdot)$. Conditioned on it, $E_{\rm obs}(t)$ is a deterministic time-warp of a complex Gaussian field, so the Siegert identity holds at each fixed pair $(t, t+\tau)$ as a purely algebraic Gaussian moment identity, with no stationarity required. (Ergodicity of $E_{\rm src}$ allows the conditional time-average to be replaced by the conditional ensemble average; the subsequent average over the stationary $\delta t$ process then yields a function of $\tau$ only, by independence of $E_{\rm src}$ and $\delta t$.) With $g^{(1)}_{\rm obs}(\tau \mid \delta t) = g^{(1)}_{\rm src}(\tau - \Delta(\tau))$ and the intrinsic Siegert relation $g^{(2)}_{\rm src} - 1 = |g^{(1)}_{\rm src}|^2$:

$$g^{(2)}_{\rm obs}(\tau \mid \delta t) - 1 = g^{(2)}_{\rm src}(\tau - \Delta(\tau)) - 1. \tag{28}$$

Averaging over the jitter distribution:

$$g^{(2)}_{\rm obs}(\tau) - 1 = \left\langle g^{(2)}_{\rm src}(\tau - \Delta(\tau)) - 1 \right\rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma^2_\Delta(\tau)) \left[ g^{(2)}_{\rm src}(\tau - \Delta) - 1 \right] d\Delta. \tag{29}$$

This Gaussian kernel relies on the CLT argument of Section II C, which requires $\tau \gg \tau_{\rm foam}$; for $\tau \lesssim \tau_{\rm foam}$ the kernel may deviate from Gaussian and Eq. (29) should be regarded as approximate in that regime (see Section II B for the definition and limits of $\tau_{\rm foam}$).

### B. Narrow-Band Limit: $g^{(2)}$ is Unchanged to Leading Order

The intensity $I_{\rm obs}(t) = |E_{\rm obs}(t)|^2$ is blind to the carrier phase $e^{i\omega_0\delta t(t)}$: only the envelope matters. In the narrow-band limit $\delta t \ll \tau_c$, the jitter does not appreciably shift the envelope, $A(t - \delta t) \approx A(t)$, so $I_{\rm obs}(t) \approx |A(t)|^2 = I_{\rm src}(t)$ to leading order in $\delta t/\tau_c$. Consequently:

$$g^{(2)}_{\rm obs}(\tau) = g^{(2)}_{\rm src}(\tau) \qquad \text{(leading order in } \delta t/\tau_c), \tag{30}$$

with corrections of order $(\sigma_\ell/c\tau_c)^2$ from both envelope-timing and carrier-phase contributions to the intensity. For $\alpha = 1/2$ at $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$: $\delta t_{\rm rms}/\tau_c \approx 0.28$, so corrections are at the $\sim$8% level and the exact convolution Eq. (29) should be used for precision work. For $\alpha = 2/3$ the expansion parameter is $\sim 10^{-11}$ and Eq. (30) is exact to any practical precision.

By contrast, $g^{(1)}$ *is* sensitive to the carrier phase because $\sigma_\phi = \omega_0\sigma_\ell/c$ can be large even when $\delta t \ll \tau_c$: for the same parameters, $\sigma_\phi \approx 280 \gg 1$.

## V. SIEGERT RELATION VIOLATION

The Siegert relation $g^{(2)}(\tau) - 1 = |g^{(1)}(\tau)|^2$ holds for any stationary complex Gaussian field and provides a powerful internal consistency check: once $g^{(1)}$ is known, the expected $g^{(2)}$ is fixed. As foreshadowed in Section IV, foam violates this relation because $g^{(1)}_{\rm obs}$ is suppressed by the carrier-phase factor $\mathcal{F}_{\rm foam}$ while $g^{(2)}_{\rm obs}$ is not. This section derives and characterizes the Siegert discrepancy $\mathcal{S}(\tau) \equiv [g^{(2)}_{\rm obs} - 1] - |g^{(1)}_{\rm obs}|^2$.

### A. General Bound

Define the Siegert discrepancy:

$$\mathcal{S}(\tau) \equiv \left[ g^{(2)}_{\rm obs}(\tau) - 1 \right] - \left| g^{(1)}_{\rm obs}(\tau) \right|^2. \tag{31}$$

Let $X \equiv g^{(1)}_{\rm src}(\tau - \Delta(\tau))$, a complex random variable. From Eqs. (14) and (29):

$$g^{(1)}_{\rm obs}(\tau) = \langle X \rangle_{\delta t}, \qquad g^{(2)}_{\rm obs}(\tau) - 1 = \langle |X|^2 \rangle_{\delta t}. \tag{32}$$

Therefore

$$\mathcal{S}(\tau) = \langle |X|^2 \rangle - |\langle X \rangle|^2 \geq 0, \tag{33}$$

by the Cauchy–Schwarz inequality, with equality only when $\Delta(\tau)$ has zero variance (i.e., at $\tau = 0$ or in the absence of foam). The Siegert relation is violated for all $\tau \neq 0$ whenever foam is present. This non-negativity holds for a point source; for extended sources the cross-ray differential jitter can drive $\mathcal{S}(0) < 0$ (see Section VI). The Siegert violation is not a non-classical signature: the foam-modified field remains a classical stochastic process, and the classical inequalities $g^{(2)}(0) \geq g^{(2)}(\tau)$ and $g^{(2)}(0) > 1$ are preserved; it is only the Gaussian-statistics identity relating $g^{(2)}$ to $g^{(1)}$ that is broken.

### B. Sensitivity Scale Decoupling

The violation arises because $g^{(1)}$ and $g^{(2)}$ respond to foam on different scales. The first-order coherence is controlled by $\sigma_\phi = 2\pi\sigma_\ell(D)/\lambda$, the rms accumulated phase along a single ray; $g^{(1)}$ is sensitive to it because the carrier phase $e^{i\omega_0 \delta t}$ appears directly in the field correlation. The second-order coherence is controlled by $\sigma_\Delta(\tau)/\tau_c$, the rms differential jitter scaled by the coherence time; $g^{(2)}$ is sensitive to it because intensity correlations depend on how much the jitter *changes* between two detection events, not its absolute value. These two scales are related by

$$\frac{\sigma_\Delta(\tau)}{\tau_c} = \frac{\sqrt{2[1 - C_\alpha(\tau)]}\,\sigma_\ell(D)}{c\tau_c} = \sigma_\phi \cdot \frac{\Delta\lambda}{\lambda} \cdot \sqrt{2[1 - C_\alpha(\tau)]}, \tag{34}$$

using $\omega_0 \tau_c = \lambda/\Delta\lambda$. The factor $\Delta\lambda/\lambda \ll 1$ means that even when $\sigma_\phi \gg 1$ (fringes erased), $\sigma_\Delta(\tau)/\tau_c$ can be small so $g^{(2)}$ remains near its intrinsic value while $g^{(1)}$ is suppressed. This decoupling is the physical root of the Siegert violation, and is only possible within the narrow-band regime discussed in Section III C. Numerical values of both scales are given in Table I (evaluated at the dynamical limit $C_\alpha \to 0$; for quasi-static foam $C_\alpha \approx 1$ the ratio vanishes at all lags).

TABLE I. Sensitivity scales at $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$, $A_\alpha = 1$, $C_\alpha \to 0$. The $\alpha = 1/2$ case is observationally excluded [14]; $\alpha = 2/3$ (holographic) is the most sensitive surviving model.

| Scale | $\alpha = 1/2$ (excluded) | $\alpha = 2/3$ |
|---|---|---|
| $\sigma_\phi$ | 280 | $2.5 \times 10^{-8}$ |
| $\sigma_\Delta(\tau)/\tau_c$ | 0.40 | $3.6 \times 10^{-11}$ |

### C. Explicit Form and Properties in the Narrow-Band Limit

In the narrow-band limit, $g^{(2)}_{\rm obs} \approx g^{(2)}_{\rm src}$ (Eq. (30)), while $g^{(1)}_{\rm obs}$ is suppressed by $\mathcal{F}_{\rm foam}(\tau)$ (Eq. (22)). Using the intrinsic Siegert relation $g^{(2)}_{\rm src}(\tau) - 1 = |g^{(1)}_{\rm src}(\tau)|^2$, to leading order:

$$\mathcal{S}(\tau) \approx |g^{(1)}_{\rm src}(\tau)|^2 \Big(1 - e^{-2\sigma_\phi^2[1 - C_\alpha(\tau; D)]}\Big). \tag{35}$$

In the weak-jitter limit $\sigma_\phi \ll 1$, $\mathcal{S} \approx 2\sigma_\phi^2[1 - C_\alpha]\,|g^{(1)}_{\rm src}|^2$ linear in $\sigma_\phi^2$ and small. In the strong-jitter limit $\sigma_\phi \gg 1$, the exponential vanishes and $\mathcal{S} \approx |g^{(1)}_{\rm src}(\tau)|^2$: $g^{(2)}_{\rm obs} - 1$ retains its intrinsic value while $|g^{(1)}_{\rm obs}|^2 \to 0$, producing a full violation. Quasi-static foam ($C_\alpha \approx 1$) gives $\mathcal{S} \approx 0$ at all lags foam is invisible to all coherence observables in this limit. The discrepancy $\mathcal{S}(\tau)$ is a direct observable both $g^{(1)}_{\rm obs}$ and $g^{(2)}_{\rm obs}$ are measurable from the same dataset and the modulation beyond the prefactor $|g^{(1)}_{\rm src}(\tau)|^2$ is controlled solely by $\alpha$, $D$, and $C_\alpha(\tau; D)$, independently of the unknown source spectrum.

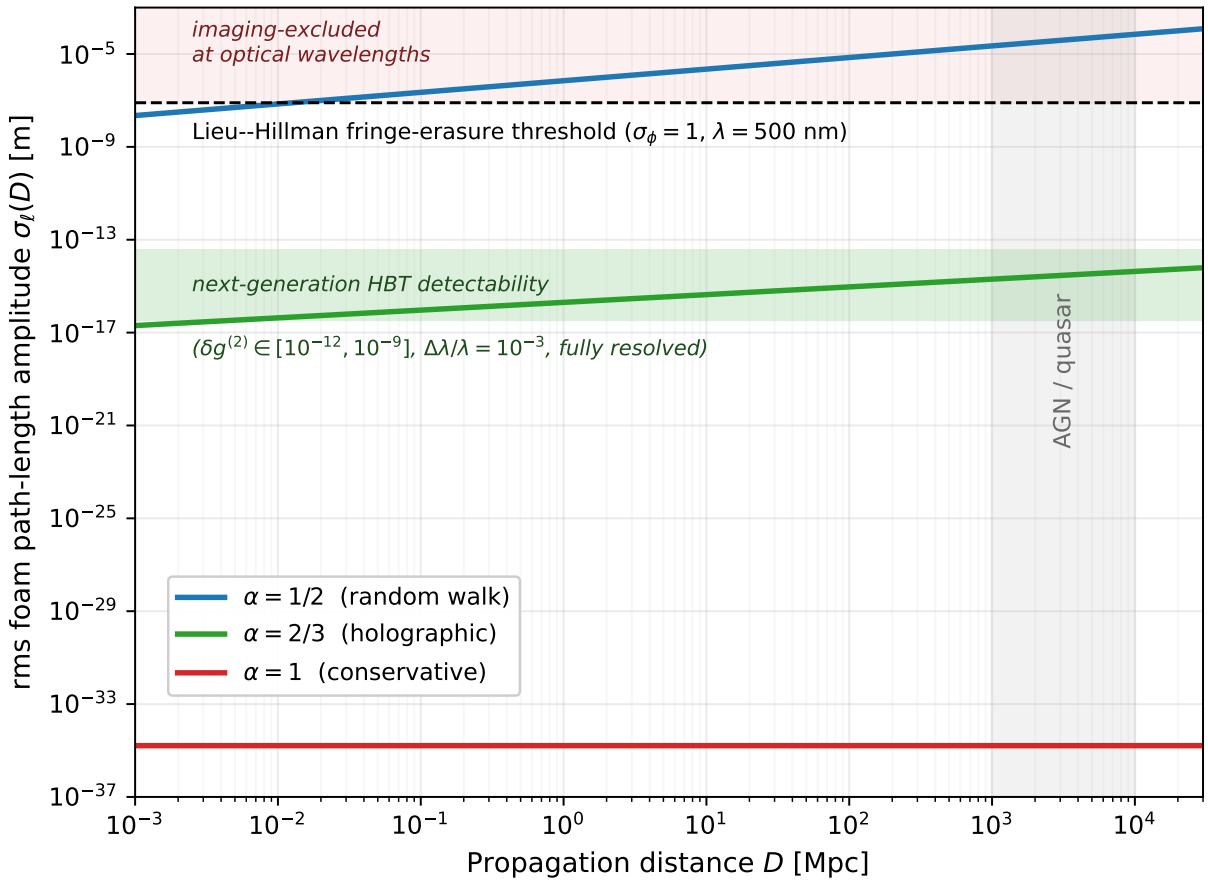


FIG. 1. Foam-induced rms path-length amplitude $\sigma_\ell(D)$ for the three Ng–Perlman exponents (Eq. (4) with $A_\alpha = 1$), shown against the principal optical sensitivity thresholds. The horizontal dashed line is the Lieu–Hillman fringe-erasure threshold $\sigma_\ell = \lambda/(2\pi)$ at $\lambda = 500$ nm; the shaded region above it is excluded by extragalactic imaging tests [11–14]. The shaded green band is the next-generation HBT detectability range at fiducial bandwidth $\Delta\lambda/\lambda = 10^{-3}$ in the fully resolved limit ($\Phi_{\rm eff} \to 0$), corresponding to $\delta g^{(2)} \in [10^{-12}, 10^{-9}]$ between the modal shot-noise floor and the projected systematic floor. The vertical grey band marks the AGN/quasar redshift range $D \sim 1$–$10$ Gpc. The random-walk exponent $\alpha = 1/2$ crosses the imaging exclusion at $D \sim 13$ kpc and is therefore ruled out for any extragalactic target. The holographic exponent $\alpha = 2/3$ sits within the HBT detectability band over the AGN regime, indicating that the second-order observables developed in Sections V–VI approach the boundary of next-generation HBT instrumentation. The conservative exponent $\alpha = 1$ has $\sigma_\ell = \ell_{\rm P}$ regardless of $D$ and lies far below all current and projected sensitivity floors.

## VI. EXTENDED SOURCES: TRANSVERSE FOAM COHERENCE

We now generalize from a point source to an extended incoherent source. The first-order coherence $g^{(1)}_{\rm obs}$ turns out to be unchanged from the point-source result, because $g^{(1)}$ probes only single-ray statistics. The second-order coherence $g^{(2)}_{\rm obs}$, however, acquires new structure through cross-ray differential jitter between rays arriving from different sky directions, producing a suppression of the zero-lag bunching peak that directly probes the transverse foam coherence length $r_\perp$.

### A. Field Model and Spatiotemporal Covariance

We generalize to an extended incoherent source whose angular brightness distribution is $I(\boldsymbol{\theta})$ (normalized to unity). The analysis assumes a point (single spatial mode) detector; a finite aperture would introduce Van Cittert–Zernike geometric phases that modify the angular coherence. The detected field is a weighted superposition over rays labelled

by direction offset $\boldsymbol{\theta}$:

$$E_{\rm obs}(t) = \int d\boldsymbol{\theta}\, I^{1/2}(\boldsymbol{\theta})\, E_{\boldsymbol{\theta}}(t - \delta t(t;\boldsymbol{\theta}))\,, \tag{36}$$

where the $E_{\boldsymbol{\theta}}$ are independent, identically distributed complex Gaussian processes (one per sky direction), and each ray carries its own foam jitter $\delta t(t;\boldsymbol{\theta})$. The spatiotemporal covariance of the path-length aggregate is given by Eq. (6) of Section II, with $b(\boldsymbol{\theta},\boldsymbol{\theta}') \simeq D|\boldsymbol{\theta}-\boldsymbol{\theta}'|$ the transverse separation between rays evaluated at the source end. For foam distributed uniformly along the line of sight the ray separation at position $z$ is $z|\boldsymbol{\theta}-\boldsymbol{\theta}'|$, so the full cross-covariance integral involves an average over $0 \le z \le D$; setting $b = D|\boldsymbol{\theta}-\boldsymbol{\theta}'|$ evaluates the separation at the source plane. Because $r_\perp$ is treated as an effective free parameter, the $O(1)$ geometric factor from the $z$-integral is absorbed into its definition; its value is constrained empirically by the angular-size dependence of $\delta g^{(2)}$, derived in Section VI D.

### B. First-Order Coherence is Unchanged

The numerator of $g^{(1)}_{\rm obs}(\tau)$ involves cross terms $\langle E^*_{\boldsymbol{\theta}}(\ldots)\, E_{\boldsymbol{\theta}'}(\ldots)\rangle$. Because different source directions are spatially incoherent,

$$\langle E^*_{\boldsymbol{\theta}}(s)\, E_{\boldsymbol{\theta}'}(s')\rangle \propto \delta(\boldsymbol{\theta}-\boldsymbol{\theta}'), \tag{37}$$

so only the diagonal $\boldsymbol{\theta}=\boldsymbol{\theta}'$ contributes. Each diagonal term reproduces the single-ray result of Section III C, and the angular integral over $I$ gives unity. Therefore:

$$g^{(1)}_{\rm obs}(\tau) = g^{(1)}_{\rm src}(\tau)\cdot e^{-\sigma^2_\phi[1-C_\alpha(\tau;D)]}, \tag{38}$$

identical to the point-source result Eq. (22). The first-order coherence probes only single-ray statistics; the transverse foam structure leaves no imprint on $g^{(1)}$.

### C. Second-Order Coherence for an Extended Source

The unnormalized intensity correlator is $\langle I(t)I(t+\tau)\rangle = \langle E^*(t)E(t)E^*(t+\tau)E(t+\tau)\rangle$, where $E \equiv E_{\rm obs}$ is the field of Eq. (36). Expanding the double angular integral and applying the Wick (moment) theorem to the four-point function of the zero-mean complex Gaussian field $E_{\boldsymbol{\theta}}$ produces two pairings (the $I^{1/2}$ weight factors enter the angular integral and are suppressed here for clarity):

$$\langle E^*_{\boldsymbol{\theta}_1} E_{\boldsymbol{\theta}_2} E^*_{\boldsymbol{\theta}_3} E_{\boldsymbol{\theta}_4}\rangle = \langle E^*_{\boldsymbol{\theta}_1} E_{\boldsymbol{\theta}_2}\rangle\langle E^*_{\boldsymbol{\theta}_3} E_{\boldsymbol{\theta}_4}\rangle + \langle E^*_{\boldsymbol{\theta}_1} E_{\boldsymbol{\theta}_4}\rangle\langle E^*_{\boldsymbol{\theta}_3} E_{\boldsymbol{\theta}_2}\rangle. \tag{39}$$

The first pairing contributes $\langle I(t)\rangle\langle I(t+\tau)\rangle$ (the disconnected term, which cancels when forming $g^{(2)}-1$). The second pairing gives the connected contribution. Because different source directions are spatially incoherent, $\langle E^*_{\boldsymbol{\theta}}(s)\, E_{\boldsymbol{\theta}'}(s')\rangle \propto \delta(\boldsymbol{\theta}-\boldsymbol{\theta}')$, so the connected pairing requires $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_4$ and $\boldsymbol{\theta}_3 = \boldsymbol{\theta}_2$. The four-fold angular integral therefore collapses to a double integral over ray pairs $(\boldsymbol{\theta},\boldsymbol{\theta}')$, each pair contributing its own jitter-convolved $g^{(2)}_{\rm src}$:

$$g^{(2)}_{\rm obs}(\tau) - 1 = \iint d\boldsymbol{\theta}\, d\boldsymbol{\theta}'\; I(\boldsymbol{\theta})I(\boldsymbol{\theta}') \int_{-\infty}^{\infty} \mathcal{N}\big(\Delta; 0, \sigma^2_\Delta(\tau;\boldsymbol{\theta},\boldsymbol{\theta}')\big)\; \big[g^{(2)}_{\rm src}(\tau-\Delta) - 1\big]\, d\Delta, \tag{40}$$

where $\sigma^2_\Delta(\tau;\boldsymbol{\theta},\boldsymbol{\theta}')$ is from Eq. (9) (Section II C). The key new feature relative to the point-source case is that for $b \gg r_\perp$, $\Phi \to 0$ and $\sigma^2_\Delta \to 2\sigma^2_\ell/c^2$ at all lags, including $\tau = 0$. In the limit $\Phi \equiv 1$ (all rays transversely coherent), every kernel reduces to the single-ray $\mathcal{N}(0, \sigma^2_\Delta(\tau))$ and Eq. (40) reduces to the point-source result Eq. (29).

### D. Effective Transverse Coherence Factor

To reduce Eq. (40) to a tractable form, define the effective transverse coherence factor:

$$\Phi_{\rm eff} \equiv \iint d\boldsymbol{\theta}\, d\boldsymbol{\theta}'\; I(\boldsymbol{\theta})I(\boldsymbol{\theta})\, \Phi\!\left(\frac{b(\boldsymbol{\theta},\boldsymbol{\theta}')}{r_\perp}\right). \tag{41}$$

This replaces the ray-dependent differential variance with a single effective variance:

$$\sigma^2_{\tau,\mathrm{eff}}(\tau) \equiv \frac{2\sigma^2_\ell(D)}{c^2}\Big[1 - C_\alpha(\tau; D)\,\Phi_{\mathrm{eff}}\Big], \tag{42}$$

and allows Eq. (40) to be approximated by the simpler 1D convolution of Eq. (29) with $\sigma^2_\Delta$ replaced by $\sigma^2_{\tau,\mathrm{eff}}$. The two limits are: $\Phi_{\mathrm{eff}} \to 1$ (point-like source or $r_\perp \gg D\,\Delta\theta$) recovers the single-ray result; $\Phi_{\mathrm{eff}} \to 0$ (fully incoherent, $r_\perp \ll D\,\Delta\theta$) maximizes the effective variance. The approximation is exact at both limits. At intermediate angular sizes ($r_\perp \sim D\,\Delta\theta$), $\Phi(b/r_\perp)$ varies across ray pairs and replacing the full double integral by a single kernel introduces errors proportional to the variance of $\Phi(b/r_\perp)$ across the source brightness distribution. A precise bound requires knowledge of the functional form of $\Phi$ and the source profile; for the purposes of this paper the $\Phi_{\mathrm{eff}}$ reduction is used as a tractable approximation. The accuracy of this approximation at intermediate angular sizes has not been quantified in the present work; it must be assessed jointly with the CLT breakdown already noted in Section II C, since both occur in the same regime $b \sim r_\perp$.

### E. Extended-Source Siegert Discrepancy and Zero-Lag Bunching

For the extended source, $g^{(1)}_{\mathrm{obs}}(\tau)$ is unchanged from the point-source result (Eq. 38), while $g^{(2)}_{\mathrm{obs}}(\tau)$ is given by the general mixture Eq. (40). Unlike the point-source case, in the narrow-band limit $g^{(2)}_{\mathrm{obs}}$ is *not* equal to $g^{(2)}_{\mathrm{src}}$: the cross-ray differential jitter modifies the intensity correlation through the envelope timing, even though the carrier phase does not affect individual intensities. The Siegert discrepancy is therefore

$$\mathcal{S}(\tau) = \Big[g^{(2)}_{\mathrm{obs}}(\tau) - 1\Big] - \Big|g^{(1)}_{\mathrm{obs}}(\tau)\Big|^2, \tag{43}$$

where $g^{(2)}_{\mathrm{obs}}(\tau) - 1$ is given by Eq. (40) and $|g^{(1)}_{\mathrm{obs}}(\tau)|^2 = |g^{(1)}_{\mathrm{src}}(\tau)|^2\, e^{-2\sigma^2_\phi[1-C_\alpha(\tau)]}$.

At $\tau = 0$, $g^{(1)}_{\mathrm{obs}}(0) = 1$ exactly by the spatial-incoherence argument of Section VI, so $\mathcal{S}(0) = g^{(2)}_{\mathrm{obs}}(0) - 2$. Evaluating the zero-lag effective variance with $C_\alpha(0; D) = 1$ gives

$$\sigma^2_0 \equiv \frac{2\sigma^2_\ell(D)}{c^2}(1 - \Phi_{\mathrm{eff}}), \tag{44}$$

which is strictly positive whenever $\Phi_{\mathrm{eff}} < 1$, so the ray-pair mixture reduces to a single Gaussian kernel of width $\sigma_0$.

For a Lorentzian source, $g^{(1)}_{\mathrm{src}}(\tau) = e^{-|\tau|/\tau_c}$, so the Siegert relation gives $g^{(2)}_{\mathrm{src}}(\tau) - 1 = e^{-2|\tau|/\tau_c}$. The zero-lag convolution over $\Delta \sim \mathcal{N}(0, \sigma^2_0)$ then becomes, setting $s \equiv \sigma_0/\tau_c$ and splitting on the sign of $\Delta$ (both halves contributing equally by symmetry of $|\Delta|$),

$$\left\langle e^{-2|\Delta|/\tau_c}\right\rangle = \frac{2}{\sqrt{2\pi}\,\sigma_0}\int_0^\infty e^{-2\Delta/\tau_c} e^{-\Delta^2/2\sigma^2_0} d\Delta = e^{2s^2}\,\mathrm{erfc}\Big(\sqrt{2}\,s\Big), \tag{45}$$

which gives 1 as $s \to 0$ (no suppression) and $\to 0$ as $s \to \infty$ via the standard asymptotic $e^{x^2}\mathrm{erfc}(x) \sim 1/(x\sqrt{\pi})$ (full suppression, $g^{(2)}_{\mathrm{obs}}(0) \to 1$). So

$$g^{(2)}_{\mathrm{obs}}(0) = 1 + e^{2s^2}\,\mathrm{erfc}\Big(\sqrt{2}\,s\Big), \qquad s \equiv \sigma_0/\tau_c, \tag{46}$$

and the zero-lag bunching peak is suppressed below its ideal value of 2 by

$$\delta g^{(2)} \equiv 2 - g^{(2)}_{\mathrm{obs}}(0) = 1 - e^{2s^2}\,\mathrm{erfc}\Big(\sqrt{2}\,s\Big), \tag{47}$$

giving

$$\mathcal{S}(0) = -\,\delta g^{(2)} \le 0. \tag{48}$$

This sign is qualitatively distinct from the point-source result $\mathcal{S}(\tau \neq 0) \ge 0$ of Section V: a negative zero-lag Siegert discrepancy is therefore a sign-flip diagnostic that distinguishes spatially resolved from unresolved foam signatures. In the point-source limit $\Phi_{\mathrm{eff}} \to 1$, $\sigma_0 \to 0$, $\delta g^{(2)} \to 0$, and $\mathcal{S}(0) \to 0$, recovering the point-source result. Since $\sigma^2_0 \propto (1 - \Phi_{\mathrm{eff}})$ and $\Phi_{\mathrm{eff}}$ decreases as the source angular diameter increases at fixed $D$ and $r_\perp$, more extended sources experience greater suppression of the bunching peak and a more negative Siegert discrepancy at zero lag.

For an astrophysically more representative Voigt profile — a Lorentzian of width $\tau_L$ convolved with a Gaussian of width $\tau_G$ (modelling natural plus Doppler/turbulent broadening) — the coherence function is $g^{(1)}_{\rm src}(\tau) = e^{-|\tau|/\tau_L}\, e^{-\tau^2/2\tau_G^2}$, giving $g^{(2)}_{\rm src}(\tau) - 1 = e^{-2|\tau|/\tau_L}\, e^{-\tau^2/\tau_G^2}$. The zero-lag convolution over $\Delta \sim \mathcal{N}(0, \sigma_0^2)$ combines the source Gaussian with the jitter kernel into an effective width

$$\tilde{\sigma}^2 = \frac{\sigma_0^2\, \tau_G^2}{2\sigma_0^2 + \tau_G^2}, \tag{49}$$

yielding

$$g^{(2)}_{\rm obs}(0) - 1 = \frac{\tilde{\sigma}}{\sigma_0}\, e^{2\tilde{s}^2}\, \mathrm{erfc}\Big(\sqrt{2}\,\tilde{s}\Big), \qquad \tilde{s} \equiv \tilde{\sigma}/\tau_L. \tag{50}$$

In the limit $\tau_G \to \infty$ (pure Lorentzian), $\tilde{\sigma} \to \sigma_0$ and Eq. (50) reduces to Eq. (46). In the limit $\tau_L \to \infty$ (pure Gaussian), $\tilde{s} \to 0$ and $g^{(2)}_{\rm obs}(0) - 1 = \tilde{\sigma}/\sigma_0 = \tau_G/\sqrt{2\sigma_0^2 + \tau_G^2}$, with no O(s) correction.

In the narrow-band limit $\sigma_\ell/(c\tau_c) \ll 1$ (Section III C), $\sigma_0 \ll \tau_c$ for any $\Phi_{\rm eff}$. Substituting $u = \Delta/\sigma_0$ and expanding $g^{(2)}_{\rm src}(\sigma_0 u) - 1$ for small $\sigma_0$, the convolution integral yields to first order in $s = \sigma_0/\tau_c$:

$$g^{(2)}_{\rm obs}(0) - 1 \approx \big[g^{(2)}_{\rm src}(0) - 1\big] + \sqrt{\tfrac{2}{\pi}}\, \sigma_0\, g^{(2)\prime}_{\rm src}(0^+), \tag{51}$$

where $g^{(2)\prime}_{\rm src}(0^+)$ denotes the right-derivative of $g^{(2)}_{\rm src}$ at zero lag (the function is non-differentiable at $\tau = 0$ for cusped profiles such as the Lorentzian, so the symmetric derivative does not exist there). For sources with a smooth (even) profile the right-derivative vanishes and the leading correction is $O(\sigma_0^2)$; for a Lorentzian, $g^{(2)\prime}_{\rm src}(0^+) = -2/\tau_c$, recovering $\delta g^{(2)} \approx 2\sqrt{2/\pi}\, s$, consistent with the small-$s$ expansion of Eq. (47). For a Voigt profile, the same expansion gives $\delta g^{(2)} \approx 2\sqrt{2/\pi}\, \sigma_0/\tau_L + (\sigma_0/\tau_G)^2 + \cdots$: the O(s) term is controlled solely by the Lorentzian component $\tau_L$, and Gaussian broadening (Doppler or instrumental) enters only at O(s$^2$). The O(s) scaling therefore survives whenever the line has any Lorentzian component, provided the line is spectrally resolved at the scale of $\tau_L$. This is consistent with the sensitivity-scale hierarchy of Section V B: the zero-lag suppression is small in the narrow-band regime. Observationally, this implies that a spectrograph transmitting a nearly flat bandpass (e.g. continuum), whose $g^{(2)}_{\rm src}$ is smooth at zero lag, suppresses the first-order contribution, leaving only an $O(s^2)$ residual. Maximizing foam sensitivity in $\delta g^{(2)}$ therefore favours emission lines with a significant Lorentzian component (natural or pressure broadening) that is spectrally resolved by the instrument; if the line is unresolved, the observed profile is the instrumental response (approximately Gaussian), the cusp is erased, and the leading correction reverts to O(s$^2$).

The suppression $\delta g^{(2)}$ arises from a purely spatial mechanism with no analogue in the point-source case: even at $\tau = 0$, photon pairs from different sky directions $\boldsymbol{\theta} \neq \boldsymbol{\theta}'$ accumulate independent instantaneous jitters $\delta t(t; \boldsymbol{\theta}) \neq \delta t(t; \boldsymbol{\theta}')$ when $b \gg r_\perp$, and their relative offset blurs the $g^{(2)}(0)$ measurement. Because $|g^{(1)}_{\rm obs}(0)|^2 = 1$ exactly is unaffected by this spatial jitter, the two observables are controlled by different mechanisms and $\mathcal{S}(0) < 0$. This does not violate any fundamental inequality: the bound $\mathcal{S} \geq 0$ established in Section V applies to a single ray, and the angular mixture of an extended source need not satisfy it.

For the region $\alpha > 1/2$ that experiments seek to probe, $\sigma_\phi \ll 1$ and the sensitivity parameter $s = \sigma_0/\tau_c = \sqrt{2(1 - \Phi_{\rm eff})}\, \sigma_\phi \cdot (\Delta\lambda/\lambda)$ is small across all realistic observational bandwidths. For the holographic model ($\alpha = 2/3$), $\sigma_\phi \sim 10^{-8}$, so that even at a high spectral resolution $R = \lambda/\Delta\lambda \sim 10^4$, one finds $s \sim 10^{-12}$. The foam signal $\delta g^{(2)}$ is therefore small throughout this regime.

Beyond raw detectability, the coherence observables serve as discriminants against conventional propagation phase screens [3]. Furthermore, for foam models with long-range transverse correlations image blurring is strongly suppressed [15], making the coherence observables developed here potentially the only channel sensitive to such models. The mathematical structure of the foam suppression factor $\mathcal{F}_{\rm foam}$ is identical to that of any Gaussian random phase screen, but two signatures distinguish foam: distance scaling and achromaticity. For spacetime foam $\sigma_\phi^2 \propto D^{2(1-\alpha)}$ ($D^1$ for $\alpha = 1/2$, $D^{2/3}$ for $\alpha = 2/3$, $D^0$ for $\alpha = 1$), whereas a thin intervening screen scales as $D$ and Kolmogorov turbulence as $D^{11/6}$; the holographic and conservative foam models predict sub-linear or flat growth qualitatively different from any known astrophysical screen. Geometrically, foam jitter ($\delta t = \Delta\ell/c$) is achromatic — independent of $\lambda$ — in contrast to plasma-based effects which scale as $\lambda^2$. The Fermi, H.E.S.S., and LHAASO time-of-flight analyses [16–18] rule out deterministic, energy-ordered LIV dispersion as a class of quantum-gravity effect, but do not constrain the stochastic achromatic foam jitter considered here (Section I); the principal chromatic alternative that the achromaticity discriminant addresses is therefore plasma propagation. In practice, the intrinsic source visibility $V_{\rm src}(\tau)$ introduces a degeneracy with foam suppression, but this is mitigated by the self-calibrating character of the

Siegert discrepancy constructed from $g^{(1)}$ and $g^{(2)}$ measurements on the same dataset. The overall source flux cancels in the ratio $g^{(2)} - 1$, but the shape of $\mathcal{S}(\tau)$ retains dependence on $|g^{(1)}_{\rm src}(\tau)|^2$; the self-calibrating property is that no separate flux calibration or source model is needed to detect a non-zero $\mathcal{S}(\tau)$, not that the source spectrum is irrelevant to its quantitative interpretation.

## VII. CONCLUSIONS

We have developed a theoretical framework for the imprint of spacetime foam, characterized by the Ng–Perlman exponent $\alpha$, on both first- and second-order optical coherence. The central finding is that foam-induced arrival-time jitter produces two classes of observable violation of relations that hold for all Gaussian chaotic sources: a deviation from the Siegert relation, $g^{(2)}(\tau) - 1 = |g^{(1)}(\tau)|^2$, and a suppression of the zero-lag bunching peak below its canonical value of $g^{(2)}(0) = 2$. These second-order observables are complementary to the fringe visibility and Strehl ratio tests based on first-order coherence that have already been applied to constrain foam models. For the holographic model ($\alpha = 2/3$), all such observables lie well within the currently inaccessible parameter space: the first-order tests probe effects at the $\sim 10^{-8}$ level, while the second-order observables are characteristically smaller at $\sim 10^{-11}$ (a recent analysis of GRB 221009A claims consistency with holographic foam blurring [19], though without formal statistical significance). The predicted signals approach but do not clearly exceed the systematic floors of $\sim 10^{-9}$–$10^{-10}$ anticipated for next-generation optical intensity interferometers, and detection would require both favorable foam parameters and significant advances in controlling detector systematics.

The advantage of the second-order approach lies not in raw sensitivity but in two distinct features: the Siegert discrepancy is self-calibrating — any deviation of $\mathcal{S}(\tau) = g^{(2)}(\tau) - 1 - |g^{(1)}(\tau)|^2$ from zero is unambiguously non-Gaussian-chaotic, requiring no separate flux calibration or source model to detect; and for extended sources whose line profile has a Lorentzian component (natural or pressure broadening), the zero-lag bunching suppression $\delta g^{(2)}$ scales linearly with $s = \sigma_0/\tau_L$ — a consequence of the cusp in $g^{(2)}_{\rm src}$ at zero lag — so that despite being a second-order coherence measurement, $\delta g^{(2)}$ is not doubly suppressed relative to the first-order observables.

For the fiducial parameters $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$, $A_\alpha = 1$: the $\alpha = 1/2$ case gives $\sigma_\phi \approx 280$ (fringes completely erased) and is already excluded by fringe-visibility measurements [12–14]. For the holographic model $\alpha = 2/3$, $\sigma_\phi \sim 10^{-8}$, so temporal point-source observables are negligibly small with current instrumentation. However, in the fully resolved limit $\Phi_{\rm eff} \to 0$ (corresponding to $D\Delta\theta \gg r_\perp$), the extended-source zero-lag bunching suppression scales linearly rather than quadratically with the small parameter $s = \sigma_0/\tau_c$ and yields $\delta g^{(2)} \approx 5.7 \times 10^{-11}$, evaluated in the dynamical limit $C_\alpha \to 0$ (a conditional upper bound). Since $s \propto \Delta\lambda/\lambda$, broader-band observations would increase $\delta g^{(2)}$ proportionally; for the holographic model the narrow-band validity condition $\sigma_\phi \cdot (\Delta\lambda/\lambda) \ll 1$ is trivially satisfied at any physically realizable bandwidth, so there is no approximation barrier to exploiting a broader bandpass, though a controlled calculation in the broadband regime — where the carrier/envelope decomposition underlying the Siegert violation is no longer rigorous — remains an open problem.

In any intensity interferometry measurement, a finite number $M$ of collected spatiotemporal modes contributes a shot-noise floor $\delta g^{(2)}_{\rm floor} \sim M^{-1}$, and the foam detectability condition $\sigma_0/\tau_c \gtrsim M^{-1}$ should be quoted alongside any observational upper bound on $\sigma_\ell(D)$ derived from a non-detection.

### A. Connection to Existing Radio-Astronomy Pipelines

The $g^{(2)}(0)$ observable developed here is already routinely computed in radio astronomy under different names: spectral kurtosis [24, 25], used for real-time RFI excision at the VLA, GBT, LOFAR, and CHIME, is constructed to estimate $g^{(2)}(0) - 1$ from channelized power samples (returning unity in expectation for a stationary Gaussian field, up to finite-sample bias corrections), as does the modulation index $m^2 \equiv {\rm Var}(I)/\langle I\rangle^2$, which equals $g^{(2)}(0) - 1$ for stationary, noise-corrected intensity samples and is used to characterize single-pulse intensity statistics of pulsars and FRBs [26]. The Siegert-violation framework developed here therefore translates directly to existing radio data products. The achievable sensitivity at radio wavelengths is set by the same combination $s \propto \sigma_\ell(D)\,\Delta\nu/c$ that controls the optical case, but with a modal shot-noise floor and systematic floor (RFI residuals, digitizer nonlinearity).

### B. Cosmological Evolution and Future Directions

A deeper question left open by the present framework is whether spacetime foam should be treated as a static medium or whether its statistical properties evolve on cosmological timescales. Throughout this paper we have

treated $\sigma_\ell^2(D)$ as a fixed parameter, an assumption justified by the negligible fractional change $H_0 T \sim 10^{-10}$ in the propagation distance over any realistic observation duration. Over cosmological lookback times, however, the foam may track the expansion of the universe in a non-trivial way. If the Ng–Perlman exponent $\alpha$ or the coefficient $A_\alpha$ itself carries a redshift dependence, then observations of sources at different redshifts would no longer trace a simple power-law $\sigma_\phi(D)$, and the distance-scaling discriminant of Section VI would need to be generalized. More speculatively, if the foam undergoes a phase transition or qualitative change in its structure at some epoch — analogous to other symmetry-breaking transitions in the early universe — this could in principle be imprinted on the coherence of light from sources bracketing that epoch, providing a novel cosmological probe. We flag these possibilities as motivation for developing a fully non-stationary, redshift-resolved treatment of foam coherence in future work.

## ACKNOWLEDGMENTS

This work was supported in part by the Director, Office of Science, Office of High Energy Physics of the U.S. Department of Energy under Contract No. DE-AC02-05CH11231.

The authors used Claude (Anthropic) and Valency Systems to assist with drafting and editing portions of this manuscript. All content was reviewed and verified by the authors, who take full responsibility for the work.

[1] J. A. Wheeler, Physical Review **97**, 511 (1955).
[2] S. W. Hawking, Nuclear Physics B **144**, 349 (1978).
[3] S. Carlip, Rep. Prog. Phys. **86**, 066001 (2023), arXiv:2209.14282 [gr-qc].
[4] Y. J. Ng and E. S. Perlman, Universe **8**, 382 (2022).
[5] Y. J. Ng, Physical Review Letters **86**, 2946 (2001).
[6] Y. J. Ng, Modern Physics Letters A **18**, 1073 (2003).
[7] A. Addazi *et al.*, Prog. Part. Nucl. Phys. **125**, 103948 (2022).
[8] R. Alves Batista *et al.*, Class. Quant. Grav. **42**, 032001 (2025), arXiv:2312.00409 [gr-qc].
[9] E. P. Verlinde and K. M. Zurek, Phys. Lett. B **822**, 136663 (2021), arXiv:1902.08207 [gr-qc].
[10] M. Lagouvardos and C. Anastopoulos, Class. Quant. Grav. **38**, 115012 (2021), arXiv:2011.08270 [gr-qc].
[11] R. Lieu and L. W. Hillman, The Astrophysical Journal **585**, L77 (2003).
[12] R. Ragazzoni, M. Turatto, and W. Gaessler, ApJ **587**, L1 (2003), arXiv:astro-ph/0303043 [astro-ph].
[13] W. A. Christiansen, Y. J. Ng, D. J. E. Floyd, and E. S. Perlman, Physical Review D **83**, 084003 (2011).
[14] E. S. Perlman, S. A. Rappaport, W. A. Christiansen, Y. J. Ng, J. DeVore, and D. Pooley, The Astrophysical Journal **805**, 10 (2015).
[15] V. S. H. Lee, K. M. Zurek, and Y. Chen, Physical Review D **109**, 084005 (2024), arXiv:2312.06757 [gr-qc].
[16] A. A. Abdo, M. Ackermann, M. Ajello, *et al.* (Fermi GBM/LAT Collaboration), Nature **462**, 331 (2009), arXiv:0908.1832 [astro-ph].
[17] F. Aharonian *et al.* (H.E.S.S. Collaboration), Phys. Rev. Lett. **101**, 170402 (2008), arXiv:0810.3475 [astro-ph].
[18] LHAASO Collaboration, Physical Review Letters **133**, 071501 (2024), arXiv:2402.06009.
[19] E. Steinbring and Y. J. Ng, arXiv e-prints (2026), arXiv:2603.18277.
[20] R. H. Brown and R. Q. Twiss, Nature **177**, 27 (1956).
[21] A. J. F. Siegert, *On the fluctuations in signals returned by many independently moving scatterers*, Tech. Rep. 465 (MIT Radiation Laboratory, 1943).
[22] A. U. Abeysekara *et al.*, Nature Astronomy **4**, 1164 (2020).
[23] A. Acharyya *et al.*, The Astrophysical Journal **966**, 28 (2024).
[24] G. M. Nita, D. E. Gary, Z. Liu, G. J. Hurford, and S. M. White, Publications of the Astronomical Society of the Pacific **119**, 805 (2007).
[25] G. M. Nita and D. E. Gary, Monthly Notices of the Royal Astronomical Society **406**, L60 (2010).
[26] P. Weltevrede, R. T. Edwards, and B. W. Stappers, Astronomy and Astrophysics **445**, 243 (2006).